

Fractional SPDE with Time-independent Noise: Solvability and Exact Asymptotics*

Le Chen[†] and Nick Eisenberg[‡]

Friday 11th June, 2021

Abstract

In this article, we study the

MSC 2010: Primary 60H15; Secondary 60H07, 37H15

Keywords: Stochastic partial differential equations; *Caputo* derivatives; *Riemann-Liouville* fractional integral; fractional Laplacian; *Malliavin* calculus; *Skorohod* integral; exact moment asymptotics; time-independent Gaussian noise.

Contents

1	Introduction	1
2	Some preliminaries	6
2.1	Malliavin Calculus	6
2.2	Some large deviation results	8
3	Existence and uniqueness of the solution	11
4	Upper bound of the asymptotics	15
5	Lower bound of the asymptotics	19
6	Appendix	26
	Bibliography	28

1 Introduction

In this paper we study the following fractional *stochastic partial differential equation* (SPDE):

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (1.1) \quad \boxed{\text{E:SPDE}}$$

*Department of Mathematics and Statistics, Auburn University, Auburn, Alabama.

[†]Email: lzc0090@auburn.edu

[‡]Corresponding author. Email: ...@auburn.edu

where $a \in (0, 2)$, $b \in (0, 2)$, $r \geq 0$, $\nu > 0$ and $\theta > 0$. Here the noise $W = \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$ is a centered and time-independent Gaussian process, defined on a complete probability space $(\Omega, \mathcal{F}, P)$, with mean zero and covariance

$$\mathbb{E}[W(\varphi)W(\psi)] = \int_{\mathbb{R}^d} \mathcal{F}\varphi(\xi)\overline{\mathcal{F}\psi(\xi)}\mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}},$$

where μ refers to the *spectral measure*, which is assumed to be a nonnegative and nonnegative definite tempered measure on $\mathbb{R}^d$. Let γ be the Fourier transform of μ , which is also a nonnegative and nonnegative measure on $\mathbb{R}^d$ thanks to Bochner's theorem. Throughout of the paper, we use $\mathcal{F}\varphi(\xi) = \int_{\mathbb{R}^d} \exp(-ix\xi)\varphi(x)dx$ to denote the Fourier transform of a test function φ .

In (1.1), $(-\Delta)^{a/2}$ refers to the *fractional Laplacian* of order a , ∂_t^b denotes the *Caputo fractional differential operator*

$$\partial_t^b f(t) := \begin{cases} \frac{1}{\Gamma(m-b)} \int_0^t d\tau \frac{f^m(\tau)}{(t-\tau)^{b+1-m}} & \text{if } m-1 < b < m, \\ \frac{d^m}{dt^m} f(t) & \text{if } b = m, \end{cases}$$

where m is an integer, and I_t^r refers to the *Riemann-Liouville fractional integral* of order $r > 0$

$$I_t^r f(t) := \frac{1}{\Gamma(r)} \int_0^t (t-s)^{r-1} f(s) ds, \quad \text{for } t > 0,$$

with the convention that when $r = 0$, $I_t^0 = \text{Id}$ reduces to the identity operator. The fundamental solutions to (1.1) consist of a triplet:

$$\left\{ Z_{a,b,r,\nu,d}(t, x), Z_{a,b,r,\nu,d}^*(t, x), Y_{a,b,r,\nu,d}(t, x) : t > 0, x \in \mathbb{R}^d \right\}.$$

Each member of the triplet can be expressed explicitly in terms of the *Fox H-function* which is much more complicated than the fundamental function for either the heat or wave equation; See Theorem 4.1 of [5]. Since we are interested in the constant one initial condition (and zero initial velocity when $b > 1$), Theorem 4.1 (*ibid.*) implies that the corresponding solution to the homogeneous equation (i.e. the solution when there is no driving source) is equal to the constant one. Hence, by setting $G(t, x) := Y_{a,b,r,\nu,d}(t, x)$, (1.1) can be written as the following integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds, \quad (1.2)$$

E:spdeint

where the stochastic integral is in the *Skorohod* sense; see Definition 2.1 below. In the following, the fundamental solution will exclusively refer to $G(t, x)$, which is indeed a smooth function for $x \neq 0$. Our results rely on the following assumption for the nonnegativity of $G(t, x)$, which is a nontrivial fact; see Remark 1.9 for sufficient conditions.

Nonnegative

Assumption 1.1 (Nonnegativity). Assume that the fundamental solution $G(t, x)$ is nonnegative for all $t > 0$ and $x \in \mathbb{R}^d$.

Regarding the noise, we formulate the following assumption in order to cover the Riesz kernel case, the fractional noise and a mixture of them:

A:Noise

Assumption 1.2 (Noise). Let $k \in \{1, \dots, d\}$ and partition the d -coordinates of $x = (x_1, \dots, x_d)$ into k distinct groups of size d_i so that $d_1 + \dots + d_k = d$. Denote $x_{(i)} = (x_{i_1}, \dots, x_{i_{d_i}})$ to be the coordinates in the i^{th} partition. Assume that the correlation function of the Gaussian noise is given by

$$\gamma(x) = \prod_{i=1}^k |x_{(i)}|^{-\alpha_i} \quad \text{with } \alpha_i \in (0, d_i). \quad (1.3) \quad \text{E:corfuncti}$$

Define $\alpha := \sum_{i=1}^k \alpha_i$.

In a recent work by Balan *et al* [1], the same equation as (1.1), but exclusively for the *stochastic wave equation* (SWE), namely, the case when $a = b = \nu = 2$ and $r = 0$, has been studied, where both the well-posedness and the exact moment asymptotics have been obtained. The corresponding *stochastic heat equation* (SHE), namely, the case when $a = 2$, $b = \nu = 1$ and $r = 0$, has been earlier studied by Hu [9], but only for the well-posedness and exclusively for the fractional noise. The corresponding moment asymptotics have been conjectured in Remark 1.9 of Balan *et al* [1] by *formally* setting $H_0 = 1$ in X. Chen [6]. Note that in X. Chen (*ibid.*), H_0 is strictly in between the open interval $(1/2, 1)$. The case $H_0 = 1$ requires a separate treatment. In this paper, by working on a more general class of SPDEs, we confirm this conjecture on the asymptotics for SHE. Moreover, our results interpolate the asymptotics for both SWE and SHE; see Remark 1.8 below.

The moment asymptotics obtained by X. Chen, such as those in [6, 7], rely heavily on the Feynman-Kac representation of the moments of the solution. However, whenever $b \neq 1$, especially for the case when $b \in (1, 2)$, we are not aware of any such Feynman-Kac formula for the moments. Instead, in the recent work by Balan *et al* [1], this difficulty has been overcome by studying the Wiener chaos expansion of the solution. In this paper, we follow the same strategy laid out by Balan *et al* (*ibid.*).

Now let us state our main results of the paper. The first main result deals with the well-posedness of the SPDE (1.1) (or (1.2)) as stated in the following theorem:

T:solvable

Theorem 1.3 (Solvability). Assume that both Assumptions 1.1 and 1.2 hold.

(1) (1.1) has a unique solution $u(t, x)$ in $L^p(\Omega)$ for all $p \geq 2$, $t > 0$, and $x \in \mathbb{R}^d$ provided that

$$0 < \alpha < \min \left(\frac{a}{b} [2(b+r) - 1], 2a, d \right). \quad (1.4) \quad \text{E:alpha}$$

(2) Otherwise, if

$$r \in [0, 1/2] \quad \text{and} \quad 0 < \alpha = \frac{a}{b} [2(b+r) - 1] \leq d, \quad (1.5) \quad \text{E:critical}$$

then we have the following two cases:

(2-i) For any $p \geq 2$, (1.1) has a unique solution $u(t, x)$ in $L^p(\Omega)$ for all $p \geq 2$ and $x \in \mathbb{R}^d$, but only for $t \in (0, T_p)$ where

$$T_p := \frac{1}{2 C_\mu(\nu, a)} [\theta (p-1)]^{\frac{b\alpha}{a} - 2(b+r)} \quad \text{with} \quad C_\mu(\nu, a) := \int_{(\mathbb{R}^d)} \frac{\mu(d\xi)}{(1 + \frac{\nu}{2} |\xi|^a)^2}. \quad (1.6) \quad \text{E:Tp}$$

(2-ii) For any $t > T_2$, the series (2.11) below diverges, that is, the $L^2(\Omega)$ -solution $u(t, x)$ to (1.1) does not exist whenever $t > T_2$.

The second main result of the paper is about the moment asymptotics. We need to introduce the following variational constants:

$$\mathcal{M}_a(\Theta f, \theta) := \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, f \rangle^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\} \quad \text{and} \quad \mathcal{M}_a := \mathcal{M}_a(|\cdot|^{-\alpha}, 1);$$

see Section 2.2 for more notation and details. Now we are ready to state the second main result as follows:

T:asym **Theorem 1.4.** *Under Assumptions 1.1 and 1.2, if condition (1.4) holds, then we have that*

$$\lim_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p = \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right), \quad (1.7)$$

where

$$\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1} \quad \text{and} \quad t_p^\beta := (p-1)^{\beta-1} t^\beta. \quad (1.8)$$

Proof. We prove the matching upper bound (4.1) and the lower bound (5.1) of (1.7) at the end of Sections 4 and 5 below, respectively, which together prove (1.7). $\square$

As a direct consequence of (1.7), one can send either t or p to infinity as follows:

C:Lim **Corollary 1.5.** *Under both Assumptions 1.1 and 1.2, if condition (1.4) holds, then*

(1) *For all $p \geq 2$ fixed, it holds that*

$$\lim_{t \rightarrow \infty} t^{-\beta} \log \mathbb{E}(|u(t, x)|^p) = p(p-1)^{\frac{1}{2(b+r) - \frac{b\alpha}{a} - 1}} \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right); \quad (1.9)$$

(2) *For all $t > 0$ fixed, it holds that*

$$\lim_{p \rightarrow \infty} p^{-\beta} \log \mathbb{E}(|u(t, x)|^p) = t^\beta \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \quad (1.10)$$

Before the end of this Introduction, let us make a few remarks:

R:Noise **Remark 1.6.** Note that the spectral density corresponding to (1.3) is given by

$$\mu(d\xi) = \varphi(\xi) d\xi \quad \text{with} \quad \varphi(\xi) = \prod_{i=1}^k C_{\alpha_i, d_i} |\xi_{(i)}|^{-(d_i - \alpha_i)}. \quad (1.11)$$

Moreover, one can find a nonnegative and nonnegative definite K such that $\gamma = K * K$ where ‘ $*$ ’ denotes the spatial convolution. Indeed, one can choose

$$K(x) = \prod_{i=1}^k \beta_{\alpha_i, d_i} |x_i|^{-(d_i - \alpha_i)}. \quad (1.12) \quad \boxed{\text{E:K}}$$

The two constants in both (1.11) and (1.12) are defined as

$$C_{\alpha, d} = \pi^{-d/2} 2^{-\alpha} \frac{\Gamma((d - \alpha)/2)}{\Gamma(\alpha/2)} \quad \text{and} \quad \beta_{\alpha, d} = \pi^{-d/4} \frac{\Gamma((d + \alpha)/4)}{\Gamma((d - \alpha)/4)} \sqrt{\frac{\Gamma((d - \alpha)/2)}{\Gamma(\alpha/2)}}. \quad (1.13) \quad \boxed{\text{E:noisecons}}$$

Remark 1.7. We have the following special cases: (1) The case $k = 1$ in (1.3) (or (1.11)) recovers the Riesz kernel case. In this case,

$$\gamma(x) = |x|^{-\alpha}, \quad \varphi(x) = C_{\alpha, d} |x|^{-(d - \alpha)} \quad \text{and} \quad K(x) = \beta_{\alpha, d} |x|^{-(d + \alpha)/2}.$$

(2) The case when $k = d$ in (1.3) (or (1.11)) recovers the time-independent fractional noise. The corresponding SHE with such noise was earlier studied by Hu [9]. For this noise, we have that

$$\gamma(x) = \prod_{i=1}^d |x_i|^{-\alpha_i}, \quad \varphi(\xi) = \prod_{i=1}^d C_{\alpha_i, 1} |\xi_i|^{-(1 - \alpha_i)} \quad \text{and} \quad K(x) = \prod_{i=1}^d \beta_{\alpha_i, 1} |x_i|^{-(1 + \alpha_i)/2}.$$

Remark 1.8. The asymptotics in (1.7) interpolates the corresponding results for the stochastic wave and heat equations:

(1) By setting

$$a = b = \nu = 2 \quad \text{and} \quad r = 0,$$

we have that

$$\beta = \frac{4 - \alpha}{3 - \alpha} \quad \text{and} \quad t_p = (p - 1)^{1/(4 - \alpha)} t,$$

and results in (1.7), (1.9), and (1.10) recover the corresponding results for the stochastic wave equation, namely, (1.9), (1.10), and (1.11) of [1], respectively.

(2) As for the stochastic heat equation case, we are not aware of the exact asymptotics results for SHE with time-independent noise case. However, as explained in Remark 1.9 of [1], one may apply Chen’s asymptotic results in [6] for SHE with time-dependent noise, by formally setting $H_0 = 1$, to obtain the corresponding time-independent results. In particular, by setting

$$a = b = 2, \quad \nu = 1 \quad \text{and} \quad r = 0,$$

we have that

$$\beta = \frac{4 - \alpha}{2 - \alpha} \quad \text{and} \quad t_p = (p - 1)^{2/(4 - \alpha)} t,$$

and results in (1.7) and (1.9) recover the corresponding results for the stochastic heat equation, namely, (1.16) and (1.17) of [1], respectively.

Remark 1.9. Regarding the nonnegativity assumption — Assumption 1.1, thanks to Theorem 4.6 of [5], we have the following four groups of sufficient conditions, under either group of which $G(t, \cdot)$ is nonnegative (see Fig. 1 for an illustration):

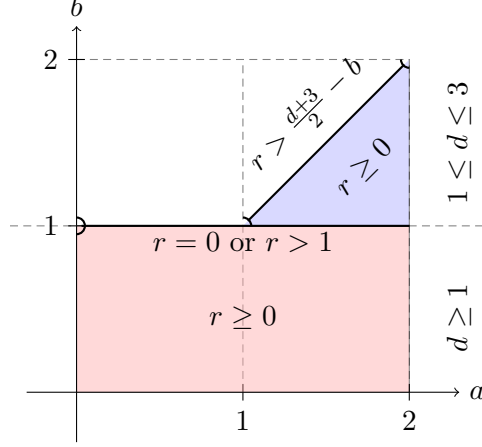


Figure 1: Illustration of the sufficient conditions (Remark 1.9) for $G(t, \cdot)$ to be nonnegative.

1. $d \geq 1$, $b \in (0, 1)$, $a \in (0, 2]$, $r \geq 0$;
2. $d \geq 1$, $b = 1$, $a \in (0, 2]$, $r = 0$ or $r > 1$;
3. $1 \leq d \leq 3$, $1 < b < a \leq 2$, $r > 0$;
4. $1 \leq d \leq 3$, $1 < b = a < 2$, $r > \frac{d+3}{2} - b$.

In the following, we first give some preliminaries in Section 2 including the Malliavin calculus, the definition of the solution, and some variational constants. The existence and uniqueness of a $L^p(\Omega)$ solution — Theorem 1.3 — is proven in Section 3. The upper bound and lower bound for (1.7) are established in Sections 4 and 5, respectively. Finally, in the appendix — Section 6, we list a few proofs of the results in the paper.

2 Some preliminaries

2.1 Malliavin Calculus

To briefly summarize [8], we start with a nonnegative and nonnegative definite tempered measure Γ with density γ in the sense that $\Gamma(dx) = \gamma(x)dx$, and

$$\int_{\mathbb{R}^d} \Gamma(dx) (\varphi * \tilde{\varphi})(x) \geq 0 \quad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}^d)$$

where $\tilde{\varphi}(x) = \varphi(-x)$. In addition we require the existence of an $r > 0$ such that

$$\int_{\mathbb{R}^d} \Gamma(dx) \frac{1}{(1 + |x|^2)^r} < \infty.$$

According to the Bochner theorem, there exists a nonnegative measure μ , often referred to as the spectral measure on $\mathbb{R}^d$ whose Fourier transform is Γ in the sense that for any $\varphi \in \mathcal{S}(\mathbb{R}^d)$

$$\int_{\mathbb{R}^d} \Gamma(dx) \varphi(x) = \int_{\mathbb{R}^d} \mu(dt) \mathcal{F}\varphi(t)$$

The functional defined on the space of test functions $\mathcal{D}(\mathbb{R}^d)$

$$C(\varphi, \psi) = \int_{\mathbb{R}} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi)$$

is nonnegative definite and thus we may associate it as the covariance functional of a zero-mean Gaussian processes, $W := \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$. In other words,

$$\mathbb{E}(W(\varphi)W(\psi)) = \int_{\mathbb{R}} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}}.$$

Let $\mathcal{H}$ be the completion of $\mathcal{D}(\mathbb{R}^d)$ with respect to $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and thus we see $\varphi \mapsto W(\varphi)$ is an isometry from the normed space $(\mathcal{D}(\mathbb{R}^d), \|\cdot\|_{\mathcal{H}})$ to $L^2(\Omega)$ where $\|f\|_{\mathcal{H}} = \langle f, f \rangle_{\mathcal{H}}$.

We denote δ the Skorohod integral with respect to W and we denote its domain $\text{Dom}(\delta)$. For any $u \in \text{Dom}(\delta)$, we write

$$\delta(u) = \int_{\mathbb{R}^d} u(x) W(dx).$$

Through construction of δ , $\delta(u)$ is a random variable in $L^2(\Omega)$. For a deeper explanation, we direct the reader to Nualart [11].

D:Sol

Definition 2.1. A random field $u = \{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$ such that $\|u(t, x)\|_2 < \infty$ for all $t \geq 0$ and $x \in \mathbb{R}^d$ is called a *mild solution* to the equation (1.1) if the random field $\{G(t-s, x-y)u(s, y) : s \in [0, t], y \in \mathbb{R}^d\}$ is Skorohod integrable for each $s \in [0, t]$ and $x \in \mathbb{R}^d$ fixed and satisfies the following integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(dy) \right) ds. \quad (2.1)$$

Through the standard Picard iteration scheme, the solution can be expressed by

$$u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)) \quad (2.2)$$

E:solwexp

where $I_k : \mathcal{H}^{\otimes k}((\mathbb{R}^d)^k) \rightarrow \mathcal{H}_k$ is the k^{th} order Skorohod integral and $\mathcal{H}_k$ is the k^{th} Wiener chaos space. The kernels $f_k(\cdot, x, t) \in \mathcal{H}^{\otimes k}((\mathbb{R}^d)^k)$ are calculated through the Piccard iteration scheme to be

$$\begin{aligned} f_n(x_1, \dots, x_k; x, t) &= \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} G(t-t_n, x-x_n) \cdots G(t_2-t_1, x_2-x_1) dt_1 \cdots dt_n \\ &= \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} G(t_1, x-x_n) \cdots G(t_n-t_{n-1}, x_2-x_1) dt_1 \cdots dt_n. \end{aligned} \quad (2.3)$$

E:wker

For ease of notation throughout this article, we write the above integrals as

$$\begin{aligned} f_n(x_1, \dots, x_n; x, t) &= \int_{[0, t]_{<}^n} G(t-t_n, x-x_n) \cdots G(t_2-t_1, x_2-x_1) d\vec{t} \\ &= \int_{[0, t]_{<}^n} G(t_1, x-x_n) \cdots G(t_n-t_{n-1}, x_2-x_1) d\vec{t}. \end{aligned}$$

where in general $d\vec{z} = \prod_{k=1}^d dz_k$ and the dimension will always be easily understood. We denote by $\widetilde{f_n}(\cdot, x, t)$ to be the symmetrization of $f_n(\cdot, x, t)$ in the variables $x_1, \dots, x_n$:

$$\widetilde{f_n}(\cdot; x, t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} f_n(x_{\rho(1)}, \dots, x_{\rho(n)}) \quad (2.4)$$

$$= \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0, t]_{<}^n} G(t - t_n, x - x_{\rho(n)}) \cdots G(t_2 - t_1, x_{\rho(2)} - x_{\rho(1)}) d\vec{t} \quad (2.5)$$

where Σ_n is the set of all permutations of $\{1, \dots, n\}$. In particular, for $x = 0$ we have

$$\widetilde{f}_n(\cdot; 0, t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0, t]_{<}^n} G(t_1, x_{\rho(1)}) \cdots G(t_n - t_{n-1}, x_{\rho(n)} - x_{\rho(n-1)}) d\vec{t}. \quad (2.6)$$

By setting $t_{n+1} = t$, the Fourier transform of the kernels, f_n , is given by

$$\mathcal{F}f_n(\cdot; x, t)(\xi_1, \dots, \xi_n) = e^{-i(\sum_{j=1}^n \xi_j) \cdot x} \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(t_{t_{k+1}} - t_k, \cdot)} \left(\sum_{j=1}^k \xi_j \right) dt. \quad (2.7)$$

Recall the notation above that $G(t, x) = Y_{a,b,r,d}(t, x)$. The following result shows an important scaling property that both $\mathcal{F}G(t, \cdot)(\xi)$ and $\left\| \widetilde{f}_n(\cdot, t, x) \right\|_{\mathcal{H}^{\otimes n}}$ have.

L:Scale

Lemma 2.2. *For any $c > 0$ and $t > 0$, the following scaling properties hold:*

$$\mathcal{F}G(t, \cdot)(c\xi) = c^{-\frac{a}{b}(b+r-1)} \mathcal{F}G\left(c^{\frac{a}{b}}t, \cdot\right)(\xi) \quad \text{and} \quad \mathcal{F}G(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}G(t, \cdot)(c^{b/a}\xi), \quad (2.8)$$

$$\mathcal{F}\widetilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = c^{n(b+r)} \mathcal{F}\widetilde{f}_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n), \quad (2.9)$$

$$\left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{[2(b+r) - b\alpha/a]n} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2. \quad (2.10)$$

Proof. The scaling properties in (2.8) is a direct consequence of the explicit expression of $\mathcal{F}G(t, \cdot)(\xi)$ as in (4.8) of [5]. Property (2.9) is an easy exercise of change of variables. Property (2.10) is a direct consequence of (2.8), (2.9), and the scaling property of the spectral measure μ . We leave the details for the interested readers. $\square$

Recall the following standard result about the existence and uniqueness of the solution when the solution can be written as the Wiener chaos expansion as in (2.2).

Theorem 2.3. *Fix any $T \in (0, \infty]$. Suppose that $f_n(\cdot, x, t) \in \mathcal{H}^{\otimes n}$ for any $t \in (0, T)$, $x \in \mathbb{R}^d$ and $n \geq 1$. Then equation (1.1) has a unique $L^2(\Omega)$ solution on $(0, T) \times \mathbb{R}^d$ if and only if the series (2.2) converges in $L^2(\Omega)$ for any $(t, x) \in (0, T) \times \mathbb{R}^d$, which is equivalent to the convergence of the series (2.11). In this case, the solution is given by (2.2) with the second moment given by*

$$\mathbb{E}(u(t, x)^2) = \sum_{n \geq 0} \theta^n n! \left\| \widetilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{for all } (t, x) \in (0, T) \times \mathbb{R}^d. \quad (2.11)$$

SS:LD

2.2 Some large deviation results

Let us first define

$$\rho := \rho_{\nu, a} = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1 + \frac{\nu}{2}|x+y|^a} \sqrt{1 + \frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx).$$

Recall that when $\mu(d\xi) = C_{\alpha,d} |\xi|^{-(d-\alpha)} d\xi$ which corresponds to the Riesz kernel $\gamma(x) = |x|^{-\alpha}$ for some $\alpha \in (0, d)$, Bass, Chen and Rosen [3] established that when $a \in (0, 2]$ and $\nu = 2$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log(\rho_{2,a}), \quad (2.12)$$

and

$$\rho_{2,a} = \Lambda_\alpha^{2-(\alpha/a)} \quad (2.13) \quad \text{E:oldrhoVar}$$

where

$$\Lambda_\alpha := \sup_{g \in \mathcal{F}_a} \left\{ \left(\int_{\mathbb{R}^{2d}} \frac{g^2(x_1)g^2(x_2)}{|x_1 + x_2|^\alpha} d\vec{x} \right)^{1/2} - \mathcal{E}_a(g, g) \right\} \quad (2.14)$$

$$= \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, |\cdot|^{-\alpha} \rangle_{L^2(\mathbb{R}^d)} - \mathcal{E}_a(g, g) \right\} \quad \text{and} \\ \mathcal{E}_a(g, g) := (2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi. \quad (2.15)$$

The following result shows the relationship between $\rho_{2,a}$ and ρ and well as the generalization of (2.12) for ρ .

L:rhoVar

Lemma 2.4. *For the Riesz kernel case, i.e., when $\gamma(x) = |x|^{-\alpha}$ for some $\alpha \in (0, d)$, it holds that*

$$\rho_{\nu,a} = \left(\frac{\nu}{2}\right)^{-\alpha/a} \Lambda_\alpha^{2-(\alpha/a)}, \quad \text{for any } \nu > 0 \text{ and } a \in (0, 2]. \quad (2.16) \quad \text{E:rhoLam}$$

In addition,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log(\rho_{\nu,a}). \quad (2.17) \quad \text{E:LogRho}$$

This lemma can be proved in the standard scaling arguments together with (2.13). For completeness, the proof is presented in the Appendix of the paper.

Next we show that ρ is finite. We do this by showing $\rho_{2,a}$ is finite and then noting that this implies that ρ is finite by 2.4. We stated in the introduction that Bass, Chen and Rosen proved in [3] that ρ is finite for the case of the Riesz kernel. We mimic his proof but were he uses the Hardy-Littlewood-Sobolev theorem, we make use of the weak Young's inequality instead.

L:rhofinite

Lemma 2.5. *For any f, g, h with $h \geq 0$ and for φ given as in equation (1.11), we have the following:*

$$\left(\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{|f(x+y)g(y)|}{\sqrt{h(x+y)}\sqrt{h(y)}} dy \right]^p \varphi(x) dx \right)^{1/p} \leq C \|f\|_2 \|g\|_2 \|h^{-1}\|_{pd/\alpha}$$

where $\alpha = \sum_{i=1}^k \alpha_i$ with $k \in \{1, \dots, d\}$, $\alpha_i \in (0, d_i)$ and $\varphi(x) = \prod_{i=1}^k |x_{(i)}|^{-(d_i - \alpha_i)}$ where $x_{(i)} = (x_{i_1}, \dots, x_{i_{d_i}})$ is the i^{th} group of $x = (x_1, \dots, x_d)$.

Proof. By observing the proof of Lemma 1.6 [3], we notice that we only need to prove equation (1.27) which says

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y)G(x)\varphi(x-y)dydx \leq C \|F\|_{2d/(d+\alpha)} \|G\|_{2d/(d+\alpha)}. \quad (2.18) \quad \text{E:fracrhofin}$$

where

$$F(x) = \frac{|f(x)|}{(h(x))^{p/2}} \quad \text{and} \quad G(x) = \frac{|g(x)|}{(h(x))^{p/2}}$$

If we can do this then the result follows by copying the rest of Chen's proof. To prove (2.18), we use the weak version of Young's inequality. Recall that weak Young's inequality [10] says

$$\left| \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} a(x)b(x-y)c(y)dx dy \right| \leq K_{p,q,r,d} \|a\|_p \|b\|_{q,w} \|c\|_r$$

where $p, q, r > 1$ and $1/p + 1/q + 1/r = 2$ and

$$\|b\|_{q,w} = \sup_A |A|^{-1/q'} \int_A |f(x)| dx$$

where A is an arbitrary Borel set of finite measure. So to prove (2.18) we wish to be able to apply this with

$$p = r = 2d/(d + \alpha) \quad \text{and} \quad q = d / \left(d - \sum_{j=1}^d \alpha_j \right) = d/(d - \alpha)$$

and, in addition, with

$$a = F, \quad c = G \quad \text{and} \quad b = \varphi$$

Again, by following the steps of the proof from the Xia Chen paper, we note that it suffices to show that $\|\varphi\|_{q,w}$ is finite with $q = d/(d - \alpha)$. Note that $1/q' = 1 - 1/q = \alpha/d$.

First we define $A_R := A_1 \times \cdots \times A_k$ where $A_i \subset \mathbb{R}^{d_i}$ and $A_i = [-R, R]$ if $d_i = 1$ and $A_i = B_R(0)$ if $d_i > 1$ where $B_R(0)$ is the ball centered at the origin with radius R . With this we have that

$$\int_{A_i} |x_{(i)}|^{-(d_i - \alpha_i)} dx_{(i)} = \begin{cases} 2 \frac{R^{\alpha_i}}{\alpha_i} & , d_i = 1 \\ |\mathbb{S}^{d_i-1}| \frac{R^{\alpha_i}}{\alpha_i} & , d_i > 1 \end{cases} \quad (2.19) \quad \boxed{\text{E:int}}$$

where we used polar coordinated to calculate the integral when $d_i > 1$ and $|\mathbb{S}^{d_i-1}|$ is the surface area of the sphere in $\mathbb{R}^{d_i}$. Also,

$$|A_i| = \begin{cases} 2R & , d_i = 1 \\ |\mathbb{S}^{d_i-1}| \frac{R^{d_i}}{d_i} & , d_i > 1. \end{cases} \quad (2.20) \quad \boxed{\text{E:leb}}$$

Recall that $1/q' = \alpha/d$. Then by combining (2.19) and (2.20) we see that

$$\begin{aligned} |A_R|^{-1/q'} \int_{A_R} \varphi(x) dx &= \left(\prod_{i=1}^k |A_i| \right)^{-1/q'} \prod_{i=1}^k \int_{A_i} |x_{(i)}|^{-(d_i - \alpha_i)} dx_{(i)} \\ &= R^{-\alpha} \left(\prod_{i=1}^k \begin{cases} 2 & , d_i = 1 \\ |\mathbb{S}^{d_i-1}|/d_i & , d_i > 1. \end{cases} \right) \times R^{\alpha} \left(\prod_{i=1}^k \begin{cases} 2/\alpha_i & , d_i = 1 \\ |\mathbb{S}^{d_i-1}|/\alpha_i & , d_i > 1 \end{cases} \right) \\ &= K \end{aligned}$$

where the constant K does not depend on R . Hence we that

$$\sup_{R>0} |A_R|^{-1/q'} \int_{A_R} \varphi(x) dx = K.$$

and by the symmetry of φ we can in fact say that

$$\sup_B |B|^{-1/q'} \int_B \varphi(x) dx = K$$

where B denotes the Borel sets of finite measure. Hence, $\varphi \in L_{q,w}(\mathbb{R}^d)$ with $q = d/(d - \alpha)$. This completes the proof of Lemma 2.5. $\square$

We finish this section by making a slight generalization of Λ_α . For this, we define

$$\mathcal{M}_a(\Theta f, \theta) = \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, f \rangle^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\}. \quad (2.21)$$

One notices that by making this generalization, we can apply Theorem 1.3 to the SPDE (1.1) driven by an arbitrary Gaussian noise with correlation function f as long as one can show that ρ is finite. By slightly modifying the proof of Lemma 2.3 [1], one can show that

$$\mathcal{M}_a(\Theta f, \theta) = \Theta^{\frac{a}{2a-\alpha}} \theta^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(f, 1). \quad (2.22)$$

Finally we note that

$$\Lambda_\alpha = \mathcal{M}_a(|\cdot|^{-\alpha}, 2) = 2^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(|\cdot|^{-\alpha}, 1) \quad (2.23)$$

and by defining $\mathcal{M}_a := \mathcal{M}_a(|\cdot|^{-\alpha}, 1)$ and by applying Theorem 2.4 we see that

$$\rho = \left(\frac{\nu}{2}\right)^{-\alpha/a} \left[2^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a\right]^{2-(\alpha/a)} = \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}. \quad (2.24) \quad \text{E:rhoMrep}$$

3 Existence and uniqueness of the solution

In this section, we will prove Theorem 1.3. The proof will need the following lemma:

Lemma 3.1 (Lemma 3.5 of [1]). *If $H : [0, \infty) \rightarrow [0, \infty)$ is a non-decreasing function, then*

$$2 \int_0^\infty e^{-2t} H^2(t) dt \leq \left(\int_0^\infty e^{-t} H(t) dt \right)^2. \quad (3.1) \quad \text{E:doubleExp}$$

The proof follows the same strategy as [1, Section 3] with minor changes such as

$$\frac{1}{1 + |\xi|^2} \quad \text{replaced by} \quad \frac{1}{1 + \frac{\nu}{2} |\xi|^a}. \quad (3.2) \quad \text{E:Changes}$$

Nevertheless, for completeness, here we streamline and reorganize this proof as follows.

Proof of Theorem 1.3. We first introduce some notation. Let $L(x)$ be the Laplace transform of $G(\cdot, x)$ evaluated at one and calculate its Fourier transform $\mathcal{F}L(\xi)$ as follows:

$$L(x) = \int_0^\infty e^{-t} G(t, x) dt \quad \text{and} \quad \mathcal{F}L(\xi) = \int_0^\infty e^{-t} \mathcal{F}G(t, \cdot)(\xi) dt = \frac{1}{1 + \frac{\nu}{2} |\xi|^a}; \quad (3.3) \quad \text{E:LandFL}$$

see the proof of Theorem 4.1 of [5] for the last equality. Similarly, let $L_n(\vec{y})$ to be the Laplace transform of $\widetilde{f}_n(\vec{y}, 0, \cdot)$ evaluated at one, namely,

$$L_n(\vec{y}) := n! \int_0^\infty e^{-t} \widetilde{f}_n(\vec{y}; 0, t) dt$$

$$= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0,t]^n} \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{s} dt$$

with the convention that $s_0 = 0$ and $y_{\sigma(0)} = 0$. By the relation of convolution and the Laplace transform (or through a change of variables), we see that

$$L_n(\vec{y}) = \sum_{\sigma \in \Sigma_n} L(y_{\sigma(1)}) L(y_{\sigma(2)} - y_{\sigma(1)}) \cdots L(y_{\sigma(n)} - y_{\sigma(n-1)}). \quad (3.4)$$

Hence, from (3.3),

$$\mathcal{F}L_n(\vec{\xi}) = \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a}. \quad (3.5) \quad \boxed{\text{E:FLfn}}$$

Moreover, define

$$\begin{aligned} H_n(t, \vec{x}) &= n! \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \widetilde{f}_n(\vec{y}; 0, t) d\vec{y} \\ &= \sum_{\sigma \in \Sigma_n} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{y} d\vec{s}. \end{aligned}$$

Under the nonnegativity assumption — Assumption 1.1, we see that for any $\vec{x} \in \mathbb{R}^{nd}$ fixed, the function $t \rightarrow H_n(t, \vec{x})$ is a non-decreasing function for $t \geq 0$. For this function, we are about to apply Lemma 3.1.

Step 1. We first compute the corresponding part to the right-hand side of (3.1). By Fubini's theorem,

$$\begin{aligned} &\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \\ &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{y} d\vec{s} dt \\ &= \int_{\mathbb{R}^{dn}} \prod_{k=1}^n K(x_k - y_k) L_n(\vec{y}) d\vec{y}. \end{aligned}$$

Then an application of the Plancherel's theorem and the fact that $K * K = \gamma$ shows that

$$\int_{(\mathbb{R}^d)^n} \left[\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right]^2 d\vec{x} = \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\vec{\xi})|^2 \mu(d\vec{\xi}); \quad (3.6) \quad \boxed{\text{E:Hneq1}}$$

one may check the proof of Lemma 3.3 of [1] for more details.

Step 2. Now we compute the corresponding part to the left-hand side of (3.1). First, using the fact that $K * K = \gamma$, we see that

$$\left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n^2(t, \vec{x}) d\vec{x}; \quad (3.7) \quad \boxed{\text{E:inteq1}}$$

one may check the proof of Lemma 3.4 of [1] for more details. By the scaling property for $\mathcal{F}\widetilde{f}_n$ in (2.9), one can show that

$$\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt; \quad (3.8) \quad \boxed{\text{E:fHn}}$$

see Appendix for the proof.

Step 3. Now we can apply the Fubini theorem and Lemma 3.1 to the function $t \rightarrow H_n(t, \vec{x})$ to see that

$$\int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt \leq \int_{\mathbb{R}^{nd}} \left[\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right]^2 d\vec{x}. \quad (3.9) \quad \text{E:doubleH}$$

Therefore, combining (3.6), (3.7), and (3.9) shows that

$$\begin{aligned} \int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &\leq \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\vec{\xi})|^2 \mu(d\vec{\xi}) \\ &=: \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} T_n(\nu, a), \end{aligned} \quad (3.10) \quad \text{E:fnTn}$$

where

$$T_n(\nu, a) = \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}). \quad (3.11) \quad \text{E:Tn}$$

By the same arguments as those of Lemma 3.6 of [1] with the replacement (3.2), we see that

$$T_n(\nu, a) \leq (n!)^2 C_\mu^n(\nu, a) \quad \text{with} \quad C_\mu(\nu, a) := \int_{(\mathbb{R}^d)} \left(\frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) < \infty. \quad (3.12) \quad \text{E:muassumpt}$$

Here, the finiteness of $C_\mu(\nu, a)$ entails one part of the conditions in (1.4):

$$\int_{(\mathbb{R}^d)} \left(\frac{1}{1 + |\xi|^a} \right)^2 \mu(d\xi) = C \int_0^\infty \frac{\rho^{\alpha-1}}{(1 + \rho^a)^2} d\rho < \infty \iff 2a - \alpha + 1 > 1. \quad (3.13) \quad \text{E:alpha2}$$

Combining (3.10) and (3.12) gives that

$$\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^n dt \leq 2^{n[2(b+r)-\frac{b\alpha}{a}]} C_\mu^n(\nu, a). \quad (3.14) \quad \text{E:LNfnineq}$$

Step 4. From the scaling property (2.10) we see that

$$\begin{aligned} \int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \int_0^\infty e^{-t[2(b+r)-\frac{b\alpha}{a}]} n dt \\ &= \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right), \end{aligned} \quad (3.15) \quad \text{E:fnOne}$$

which entails another part of the conditions in (1.4):

$$2(b+r) - \frac{b\alpha}{a} > 0. \quad (3.16) \quad \text{E:alpha3}$$

From (3.14) and (3.15), we deduce that

$$\begin{aligned} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 &= \frac{1}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt \\ &\leq \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C_\mu(\nu, a) \right)^n}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \leq C \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C_\mu(\nu, a) \right)^n}{(n!)^{2(b+r)-b\alpha/a}}, \end{aligned}$$

where the constant C depends only on the value of $2(b+r) - b\alpha/a$. Therefore, by (2.11), (2.10), and the above inequality,

$$\begin{aligned} \|u(t, x)\|_2^2 &= \sum_{n \geq 0} \theta^n n! t^{[2(b+r)-b\alpha/a]n} \left\| \widetilde{f_n}(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &\leq C \sum_{n \geq 0} \theta^n t^{[2(b+r)-b\alpha/a]n} \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C_\mu(\nu, a) \right)^n}{(n!)^{2(b+r)-1-b\alpha/a}}, \end{aligned} \quad (3.17) \quad \boxed{\text{E:2nomBd}}$$

which is finite provided that (see (1.4))

$$2(b+r) - 1 - b\alpha/a > 0. \quad (3.18) \quad \boxed{\text{E:alpha4}}$$

Finally, an application of the Minkowski inequality and the hypercontractivity (see [1, Theorem B.1]) shows that for all $p \geq 2$,

$$\begin{aligned} \|u(t, x)\|_p &\leq \sum_{n \geq 0} \theta^{n/2} (p-1)^{n/2} \sqrt{n!} \left\| \widetilde{f_n}(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}} \\ &\leq \sqrt{C} \sum_{n \geq 0} \theta^{n/2} (p-1)^{n/2} t^{[2(b+r)-b\alpha/a]n/2} \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C_\mu(\nu, a) \right)^{n/2}}{(n!)^{\frac{1}{2}[2(b+r)-1-b\alpha/a]}}. \end{aligned} \quad (3.19) \quad \boxed{\text{E:pmomBd}}$$

Therefore, under condition (1.4), (1.1) has an unique $L^p(\Omega)$ -solution $u(t, x)$ for all $p \geq 2$, $t > 0$ and $x \in \mathbb{R}^d$. This proves part (1) of Theorem 1.3.

Step 5. In this step, we will prove part (2-i) of Theorem 1.3. The critical case happens when the exponent of $n!$ in (3.19) vanishes, namely,

$$\alpha = \frac{a}{b} [2(b+r) - 1].$$

Among the three inequalities in (1.4), we also need to make sure that the minimum is achieved by $\frac{a}{b} [2(b+r) - 1]$, for which, we need to additionally require $\frac{a}{b} [2(b+r) - 1] \leq d$ and

$$\frac{a}{b} [2(b+r) - 1] \leq 2a \quad \Longleftrightarrow \quad r \in [0, 1/2].$$

Putting these conditions together gives the conditions stated in (1.5). Under condition (1.5), write the right-hand side of (3.19) as $\sum_{n \geq 0} R^{n/2}$ with

$$R = \theta(p-1) [2t C_\mu(\nu, a)]^{[2(b+r)-b\alpha/a]}.$$

Therefore, by solving

$$0 < R < 1 \quad \Longleftrightarrow \quad t < T_p,$$

one finds the expression for T_p as in (1.6). Finally, under the above condition, the $L^p(\Omega)$ -solution $u(t, x)$ exists for all $x \in \mathbb{R}^d$. This proves part (2-i) of Theorem 1.3.

Step 6. Finally, the content of part (2-ii) of Theorem 1.3 is proved in part (ii) of Lemma 4.1 below. With this, we complete the whole proof of Theorem 1.3. $\square$

4 Upper bound of the asymptotics

S:upperbd

In this section, we will the upper bound of (1.7), namely,

$$\limsup_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \leq \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \quad (4.1)$$

E:momentasy

We will first establish the corresponding result for $p = 2$ in Lemma 4.2 and then apply the hypercontractivity property given by Theorem B.1 in [1] to obtain the general case for $p \geq 2$.

L:2ineq

Lemma 4.1. Assume Assumptions 1.1 and 1.2 hold.

(i) Under condition (1.4),

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right) = \log \left(2^{2(b+r) - \frac{b\alpha}{a}} \rho \right) \quad (4.2)$$

E:2ineq-a

and

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{2(b+r) - \frac{b\alpha}{a}} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left(\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \right) + \log \rho. \quad (4.3)$$

E:2ineq-b

(ii) Under condition (1.5), let T_2 be defined as in (1.6). For all $x \in \mathbb{R}^d$ and $\theta > 0$, it holds that

$$\sum_{n \geq 0} \theta^n n! \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \begin{cases} < \infty & \text{if } t < T_2, \\ = \infty & \text{if } t > T_2. \end{cases}$$

Proof. (i) Recall the definition of T_n from above as well as (2.17) which says that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{T_n}{(n!)^2} \right] = \log \rho.$$

We already showed in the proof of Theorem 1.3 above that

$$\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)$$

and then (3.14) above shows that

$$\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} T_n}{(n!)^2}.$$

Combining this and (2.17) we see that

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{1}{n} \log \left[\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right] \\ \leq \log \left(2^{[2(b+r) - \frac{b\alpha}{a}]} \right) + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{T_n}{(n!)^2} \right) \end{aligned}$$

$$= \log \left(2^{\lfloor 2(b+r) - \frac{b\alpha}{a} \rfloor} \right) + \log(\rho)$$

which proves the upper bound for (4.2).

Now we prove the lower bound. Define for $t > 0$ and $t' > 0$

$$J_n(t, t') = \int_{\mathbb{R}^{nd}} H_n(t, x) H_n(t', x) dx$$

and recall (3.7) which says

$$\left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n(t, \vec{x})^2 d\vec{x} = \frac{1}{(n!)^2} J_n(t, t).$$

Thus $J_n(t, t)$ satisfies the scaling property

$$J_n(t, t) = t^{[2(b+r) - b\alpha/a]n} J_n(1, 1).$$

Let τ and $\tilde{\tau}$ be independent exponential random variables with mean 1. By (3.6),

$$\begin{aligned} \mathbb{E}[J_n(\tau, \tau')] &= \int_0^\infty \int_0^\infty e^{-t} e^{-\tilde{t}} J_n(t, \tilde{t}) dt d\tilde{t} = \int_{\mathbb{R}^{dn}} \left[\int_0^\infty e^{-t} H_n(t, x) dt \right]^2 dx \\ &= \int_{\mathbb{R}^{nd}} |\mathcal{F}L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \cdots \mu(d\xi_n) =: T_n. \end{aligned}$$

In addition,

$$J_n(t, t') \leq J_n(t, t)^{1/2} J_n(t', t')^{1/2} = t^{[2(b+r) - b\alpha/a]n/2} t'^{[2(b+r) - b\alpha/a]n/2} J_n(1, 1).$$

Combining all of this gives us

$$\begin{aligned} T_n &= \mathbb{E}[J_n(\tau, \tilde{\tau})] \leq \mathbb{E} \left[\tau^{[2(b+r) - b\alpha/a]n/2} \right] \mathbb{E} \left[\tilde{\tau}^{[2(b+r) - b\alpha/a]n/2} \right] J_n(1, 1) \\ &= \Gamma \left(\frac{[2(b+r) - b\alpha/a]n}{2} \right)^2 J_n(1, 1) \\ &= \Gamma \left(\frac{[2(b+r) - b\alpha/a]n}{2} \right)^2 (n!)^2 \left\| \widetilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2, \end{aligned}$$

where the last line follows from (3.7). Thus so far we have shown

$$\frac{T_n}{(n!)^2} = \frac{\left(\frac{\nu}{2}\right)^{-n\alpha/a} T'_n}{(n!)^2} \leq \Gamma \left(\frac{[2(b+r) - b\alpha/a]n}{2} \right)^2 \left\| \widetilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2. \quad (4.4) \quad \boxed{\text{E:2lowbd}}$$

Now, another application of Stirling's formula shows that

$$\Gamma \left(\frac{[2(b+r) - b\alpha/a]n}{2} \right)^2 \sim \Gamma([2(b+r) - b\alpha/a]n + 1) 2^{-[2(b+r) - b\alpha/a]n} C_n$$

where $C_n = 2^{-1}[2(b+r) - b\alpha/a]^{1/2} (2\pi n)^{1/2}$. We now infer from (4.4) that

$$c_\alpha \frac{2^{[2(b+r) - b\alpha/a]n}}{C_n} \frac{1}{(n!)^2} T_n \leq \Gamma([2(b+r) - b\alpha/a]n + 1) \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$$

where c_α is a constant depending on α . Since $\frac{1}{n} \log(C_n) \rightarrow 0$ as $n \rightarrow \infty$, we obtain the lower bound

$$\begin{aligned} \liminf_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma([2(b+r) - b\alpha/a]n + 1) \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) \\ \geq \log \left(2^{[2(b+r) - b\alpha/a]} \right) + \log(\rho) \end{aligned}$$

Thus (4.2) is proven. Lastly, (4.3) follows from (4.2) and the fact that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{\Gamma(an + 1)}{(n!)^a} \right] = a \log(a).$$

This proves part (i) of Lemma 4.1.

(ii) By applying $\alpha = \frac{a}{b} [2(b+r) - 1]$ to (3.17), we see that

$$\begin{aligned} \sum_{n \geq 0} \theta^n n! \left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 &= \sum_{n \geq 0} \theta^n n! t^{[2(b+r) - b\alpha/a]n} \left\| \widetilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &\leq C \sum_{n \geq 0} \theta^n t^{[2(b+r) - b\alpha/a]n} \frac{\left(2^{[2(b+r) - \frac{b\alpha}{a}]} C_\mu(\nu, a) \right)^n}{(n!)^{2(b+r) - 1 - b\alpha/a}} \\ &= C \sum_{n \geq 0} (2\theta t C_\mu(\nu, a))^n \end{aligned}$$

and this sum is finite if and only if

$$\theta t C_\mu(\nu, a) < 1$$

which is equivalent to

$$t < \frac{1}{2\theta C_\mu(\nu, a)} = T_2 \quad \text{for } \alpha = \frac{a}{b} [2(b+r) - 1].$$

□

L:p=2

Lemma 4.2. *Under the assumptions of Theorem 1.3, we have that*

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) = \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta (\theta \rho)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right).$$

Proof. Assuming that there is an $L^2(\Omega)$ solution $u(t, x)$ and then applying the scaling property (2.10) gives us

$$\begin{aligned} \mathbb{E}(|u(t, x)|^2) &= \sum_{n \geq 0} \theta^n (n!) \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \sum_{n \geq 0} \theta^n (n!) t^{(2(b+r) - b\alpha/a)n} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} x_n R_n t^{(2(b+r) - b\alpha/a)n} \end{aligned}$$

where

$$R_n = (n!)^{2(b+r) - b\alpha/a} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{and} \quad x_n = \frac{\theta^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}}.$$

Notice that (4.3) above says that

$$\frac{1}{n} \log(R_n) \rightarrow \log \left(\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho \right) \quad \text{as } n \rightarrow \infty.$$

Now define R to be

$$R = \left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho.$$

We want to find a β and A so that

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} x_n R^n \left(t^{2(b+r) - b\alpha/a} \right)^n = A$$

By Lemma A.3 of [1],

$$\lim_{t \rightarrow \infty} \left[\frac{1}{(\theta R) t^{2(b+r) - b\alpha/a}} \right]^{\frac{1}{2(b+r) - (b\alpha/a) - 1}} \log \sum_{n \geq 0} \frac{[(\theta R) t^{2(b+r) - b\alpha/a}]^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}} = 2(b+r) - (b\alpha/a) - 1$$

and by an easy algebraic manipulation

$$\begin{aligned} & \lim_{t \rightarrow \infty} \left[\frac{1}{t^{2(b+r) - b\alpha/a}} \right]^{\frac{1}{2(b+r) - (b\alpha/a) - 1}} \log \sum_{n \geq 0} \frac{[(\theta R) t^{2(b+r) - b\alpha/a}]^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}} \\ &= [2(b+r) - (b\alpha/a) - 1] (\theta R)^{\frac{1}{2(b+r) - (b\alpha/a) - 1}}. \end{aligned}$$

Hence,

$$\beta = \frac{2a(b+r) - b\alpha}{2a(b+r) - (b\alpha) - a} \quad \text{and} \quad A = [2(b+r) - (b\alpha/a) - 1] (\theta R)^{\frac{a}{2a(b+r) - b\alpha - a}}.$$

Finally, an application of Lemma A.2 of [1] shows that

$$\begin{aligned} & \lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) \\ &= \theta^{\frac{a}{2a(b+r) - b\alpha - a}} \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right), \end{aligned}$$

which completes the proof of Theorem 4.2. $\square$

Now we are ready to prove (4.1).

Proof of (4.1). We denote u_λ for $\lambda > 0$ to be the solution of the fractional equation with θ replaced with λ and $u = u_\theta$. Using theorem B.1 of [1], we see that for all $p \geq 2$,

$$\|u(t, 0)\|_p \leq \|u_{(p-1)\theta}(t, 0)\|_2$$

and by the scaling property

$$\|u_{(p-1)\theta}(t, 0)\|_2^2 = \sum_{n \geq 0} (n!)^n \theta^n (p-1)^n \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2$$

$$\begin{aligned}
&= \sum_{n \geq 0} (n!)^n \theta^n \left\| \tilde{f}_n \left(\cdot, 0, t(p-1)^{\frac{1}{2(b+r)-b\alpha/a}} \right) \right\|_{\mathcal{H}^{\otimes n}}^2 \\
&= \left\| u \left(t(p-1)^{\frac{1}{2(b+r)-b\alpha/a}}, 0 \right) \right\|_2^2.
\end{aligned}$$

Hence,

$$\|u(t, 0)\|_p \leq \left\| u \left(t(p-1)^{\frac{1}{2(b+r)-b\alpha/a}}, 0 \right) \right\|_2 =: \|u(t_p, 0)\|_2$$

and an application of Lemma 4.2 proves (4.1). $\square$

5 Lower bound of the asymptotics

In this section, we will prove the lower bound of (1.7), namely,

$$\begin{aligned}
\liminf_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p &\geq \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \\
&\times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right). \tag{5.1}
\end{aligned}$$

We start by defining the function $W_n(t, \phi)$ on $(0, \infty) \times L_{\mathbb{C}}^2(\mu)$ by

$$W_n(t, \phi) := \int_{[0, t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \mathcal{F}G(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds. \tag{5.2}$$

with $s_0 = 0$. We now give conditions for when W_n is well defined.

Lemma 5.1. *If the measure μ satisfies the relation in (3.12), then $W_n(t, \phi)$ is well defined and for any $d \geq 1$, $t > 0$ and $\phi \in L_{\mathbb{C}}^2(\mu)$. Moreover,*

$$\int_0^\infty e^{-t} W_n(t, \phi) dt = \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi) \dots \mu(d\xi_n). \tag{5.3}$$

Proof. The proof follows the exact arguments as those in the proof of Lemma 6.2 of [1] except that one needs to use the following Laplace transform:

$$\int_0^\infty e^{-t} \mathcal{F}G(t, \cdot)(\xi) dt = \frac{1}{1 + \frac{\nu}{2} |\xi|^a};$$

see the second equation in (3.3). $\square$

The next Proposition is a restatement of Proposition 6.3 of [1]. The proof follows the same proof as that of Proposition 6.3 of [1]. We will not repeat here.

Proposition 5.2. *For $f \in \mathcal{H}$, $t > 0$, and $p, q > 1$ with $p^{-1} + q^{-1} = 1$, it holds that*

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} (q-1) \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right| \tag{5.4}$$

and as a consequence, the series $\left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right|$ converges provided that $\|u(t, 0)\|_p < \infty$, which is the case under Theorem 1.3.

Now we are going to apply a scaling argument to (5.4) in order to put t and p together, from which we can determine the constants t_p and β defined in (1.8).

Proposition 5.3. *For $p, q > 1$ and $p^{-1} + q^{-1} = 1$ we have that*

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n \left(t_p^\beta, \mathcal{F}f \right) \right|, \quad (5.5) \quad \text{E:Wnprop2}$$

where the constants β and t_p are defined in (1.8).

Proof. From Proposition 5.2, we see that for any $f \in \mathcal{H}$, the inequality (5.4) holds. For some constants $V, W > 0$, which will be determined in this proof, let $f_\tau(x) := \tau^V f(\tau^W x)$ be a scaled version of f . It is clear that $f_\tau \in \mathcal{H}$. By some elementary scaling arguments (see the proof of the Lemma 6.4 of [1]), one can show that

$$\|f_\tau\|_{\mathcal{H}}^2 = \tau^{2(V-dW)+W\alpha} \|f\|_{\mathcal{H}}^2 \quad \text{and} \quad (5.6)$$

$$W_n(t, \mathcal{F}f_\tau) = \tau^n [V-W((d-\alpha)+\frac{a}{b}(b+r))] W_n \left(t \tau^{\frac{a}{b}W}, \mathcal{F}f \right). \quad (5.7)$$

Hence, an application of Proposition 5.2 to f_τ shows that

$$\begin{aligned} \|u(t, 0)\|_p &\geq \exp \left\{ -\frac{1}{2} (q-1) \|f_\tau\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f_\tau) \right| \\ &= \exp \left\{ -\frac{1}{2} (q-1) \tau^{2(V-dW)+W\alpha} \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} \tau^n [V-W((d-\alpha)+\frac{a}{b}(b+r))] W_n \left(t \tau^{\frac{a}{b}W}, \mathcal{F}f \right) \right|. \end{aligned}$$

Comparing the above lower bound with that of (5.5), we obtain the following two relations with three unknowns W , V and τ :

$$V - W \left((d-\alpha) + \frac{a}{b}(b+r) \right) = 0, \quad (5.8) \quad \text{E:VW-1}$$

$$(p-1)^{-1} \tau^{2(V-dW)+W\alpha} = t \tau^{\frac{a}{b}W}. \quad (5.9) \quad \text{E:VW-2}$$

Since (5.9) should hold for all $t > 0$ and $p \geq 2$, one can choose $\tau = (p-1)t$ to reduce the relation (5.9) to

$$\tau^{2(V-dW)+W\alpha} = \tau^{\frac{a}{b}W+1},$$

which then gives the following equation

$$2(V-dW) + W\alpha = 1 + \frac{a}{b}W. \quad (5.10) \quad \text{E:VW-3}$$

Now solve the linear equations (5.8) and (5.10) for W and V to see that

$$\begin{cases} W = \frac{b}{a}(\beta-1), \\ V = \left(\frac{a}{b}(b+r) - \alpha + d \right) \frac{b}{a}(\beta-1), \end{cases} \quad \text{with } \beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1}.$$

Therefore, the scaling f_τ and t_p^β should be

$$f_\tau(x) = \tau^{\left(\frac{a}{b}(b+r) - \alpha + d \right) \frac{b}{a}(\beta-1)} f \left(\tau^{\frac{b}{a}(\beta-1)} x \right) \quad \text{and} \quad t_p^\beta := t \tau^{\frac{a}{b}W} = (p-1)^{\beta-1} t^\beta,$$

respectively. This completes the proof of Proposition 5.3. $\square$

To remove the absolute value sign in Proposition 5.3, we want to identify all the $f \in \mathcal{H}$ for which $W_n(t, \mathcal{F}f)$ is nonnegative. In fact, if we consider the space

$$\mathcal{H}_+ = \{f \in \mathcal{H} : f \text{ is nonnegative and nonnegative definite}\}$$

then by Plancherel's theorem

$$W_n(t, \mathcal{F}f) = \int_{[0,t]^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n (f * \gamma)(x_k) \prod_{k=1}^n G(s_k - s_{k-1}, x_k - x_{k-1}) d\vec{x} d\vec{s} =: U_n(t, f)$$

with $s_0 = 0$ and $x_0 = 0$. Under Assumption 1.1, the Green's function for the fractional equation is nonnegative and hence under these conditions we get that $W_n(t, \mathcal{F}(f)) \geq 0$ for all the functions $f \in \mathcal{H}_+$. Hence we have that

$$0 \leq W_n(t, \mathcal{F}f) = U_n(t, f) < \infty \quad \text{for all } f \in \mathcal{H}_+ \text{ and for any of the four options in Assumption 1.1.}$$

where the finiteness comes from Proposition 5.3.

Now we define

$$W_n(t) := \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} W_n(t, \mathcal{F}(f)) \quad \text{and} \quad U_n(t) := \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} U_n(t, f)$$

and with this we see that

$$W_n(t) \geq U_n(t) \geq 0$$

P:LogU

Proposition 5.4. *If τ is an exponential random variable with mean one, then*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log(\rho^{1/2}).$$

Proof. We start by letting τ be an exponential random variable with mean one. With this,

$$\mathbb{E}(U_n(\tau)) = \int_0^\infty e^{-t} U_n(t) dt \geq \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \int_0^\infty e^{-t} U_n(t, f) dt$$

For an $f \in \mathcal{H}_+$ and $\|f\|_{\mathcal{H}} = 1$ we have that $\mathcal{F}f$ is nonnegative and nonnegative definite which further implies that $\mathcal{F}f$ is even. In addition, Lemma 5.1 gives us that

$$\int_0^\infty e^{-t} U_n(t, f) dt = \int_0^\infty e^{-t} W_n(t, \mathcal{F}f) dt = \int_{\mathbb{R}^{dn}} \prod_{k=1}^n \mathcal{F}f(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n)$$

Modifying the ideas in the proof of Theorem 1.2 [3] gives us the following identity:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int_{\mathbb{R}^{dn}} \prod_{k=1}^n (\mathcal{F}f)(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n) = \log[\rho(\mathcal{F}f)]$$

where $\rho(\cdot)$ is defined on the space of nonnegative and nonnegative definite functions and is given by

$$\rho(g) := \sup_{\|h\|_{L_2}=1} \int_{\mathbb{R}^d} g(\xi) \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta) h(\eta)}{\sqrt{1 + \frac{\nu}{2} |\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2} |\eta|^a}} d\eta \right] \mu(d\xi).$$

Note that since g is nonnegative and since μ has a nonnegative density φ then we may assume h is also nonnegative for the remainder of this proof. With this being said, we see that for any $f \in \mathcal{H}_+$ and $\|f\|_H = 1$

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(t)) \geq \log \rho(\mathcal{F}f)$$

We now need to calculate

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f)$$

Consider a nonnegative function $h \in L^2(\mathbb{R}^d)$. The function $h(\cdot)/\sqrt{1 + \frac{\nu}{2}|\cdot|^a} \in L^2(\mathbb{R}^d)$ so that $g_h(x) := (2\pi)^{d/2} \mathcal{F}^{-1} \left(\frac{h(\cdot)}{\sqrt{1 + \frac{\nu}{2}|\cdot|^a}} \right) (x)$ is well defined. **I still need to verify the details that** under these conditions, $g \in W_{1,a}(\mathbb{R}^d)$ with $\|g\|_{W_{1,a}} = 1$. Here is a sketch of the details. **we no longer need this part since we do not need g to be continuous, only need to be nonnegative and nonnegative definite.**

$$\begin{aligned} \|g\|_{W_{1,a}}^2 &= \|g\|_{L^2}^2 + (2\pi)^{-d} \sum_{k=1}^d \int_{\mathbb{R}^d} \xi_k^a |\mathcal{F}g(\xi)|^2 d\xi \\ &= (2\pi)^{-d} \int_{\mathbb{R}^d} (1 + |\xi|^a) |\mathcal{F}g(\xi)|^2 d\xi \\ &= \int_{\mathbb{R}^d} h(\xi)^2 d\xi = 1 \end{aligned}$$

where above we used the fact that $\|g\|_{L^2}^2 = (2\pi)^{-d} \|\mathcal{F}g\|_{L^2}^2$.

We need from this point on.

Now, for notation let $\tilde{h}(x) = h(-x)$. Note that

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta &= (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}g_h(\xi - \eta) \mathcal{F}g_h(-\eta) d\eta \\ &= (2\pi)^{-d} (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) \end{aligned}$$

Since $h(\cdot)$ is real valued then $\widetilde{\mathcal{F}g_h} = \overline{\mathcal{F}g_h} = \mathcal{F}\bar{g}_h$. Using this and the fact that $\mathcal{F}(fg) = (2\pi)^{-d} \mathcal{F}(f) * \mathcal{F}(g)$, we see that $(2\pi)^{-d} (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) = \mathcal{F}[|g_h|^2](\xi)$ and with this we see that

$$\begin{aligned} \rho(\mathcal{F}f) &= \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) \mathcal{F}[|g_h|^2](\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} \langle f, |g_h|^2 \rangle_{\mathcal{H}}. \end{aligned}$$

With this we see that

$$\sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \sup_{\|h\|_{L^2}=1} \langle f, |g_h|^2 \rangle_{\mathcal{H}} = \sup_{\|h\|_{L^2}=1} \langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}}^{1/2}$$

where the optimal f is chosen to be $|g_h|^2 / \| |g_h|^2 \|_{\mathcal{H}}$. Now we want to show that

$$|g_h|^2 \in \mathcal{H}_+ \tag{5.11} \quad \boxed{\text{E:GH+}}$$

which would imply that the supremum can be restricted to $\mathcal{H}_+$. If this were the case then we would have that

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \sup_{\|h\|_{L^2}=1} \langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}}^{1/2}$$

and notice that

$$\begin{aligned} \langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}}^{1/2} &= \left(\int_{\mathbb{R}^d} |\mathcal{F}[|g_h|^2](\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= (2\pi)^{-d} \left(\int_{\mathbb{R}^d} |(\mathcal{F}g_h * \tilde{\mathcal{F}}g_h)(\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= \left(\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta \right]^2 \mu(d\xi) \right)^{1/2} \\ &= \rho^{1/2} \end{aligned}$$

which finally leads us to the equality

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log(\rho^{1/2}).$$

Hence, we finish the proof by proving (5.11). First note that from the above we see that

$$\int_{\mathbb{R}^d} |\mathcal{F}|g_h|^2(\xi)|^2 \mu(d\xi) \leq \rho < \infty$$

and so $|g_h|^2 \in \mathcal{H}$. Lastly, since $\mathcal{F}|g_h|^2(\xi)$ is nonnegative for $\xi \in \mathbb{R}^d$ and for nonnegative $h \in L^2(\mathbb{R}^d)$ then $\nu(d\xi) = \mathcal{F}|g_h|^2(\xi)d\xi$ is a nonnegative tempered measure. The Bochner-Schwarz theorem then implies that $|g_h|^2$ is nonnegative and nonnegative definite. This shows that $|g_h|^2 \in \mathcal{H}_+$. This completes the proof of Proposition 5.4. $\square$

L:LogU

Lemma 5.5. *It holds that*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[(n!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} U_n(1) \right] \geq \log \left(\frac{\rho^{1/2}}{\left(\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r) \right)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}} \right). \quad (5.12) \quad \text{E:LogU}$$

Proof. Notice that by some elementary scaling arguments, we have that for all $t > 0$,

$$W_n(t) = t^{n\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} W_n(1) \quad \text{and} \quad U_n(t) = t^{n\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} U_n(1). \quad (5.13) \quad \text{E:WU}$$

Let τ be an exponential random variable with mean one. By (5.13),

$$\mathbb{E}[U_n(\tau)] = \mathbb{E} \left(\tau^{n\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} \right) U_n(1) = \Gamma \left(n\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1).$$

Then an application of Proposition 5.4 shows that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[\Gamma \left(n\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1) \right] \geq \log(\rho^{1/2}). \quad (5.14) \quad \text{E:lb1}$$

As a direct consequence of Stirling's formula, we have that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{\Gamma(an + 1)}{(n!)^a} \right) = a \log(a), \quad \text{for all } a > 0,$$

with which one immediately sees that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{\Gamma \left(n \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) + 1 \right)}{(n!)^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right)}} \right) = \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) \log \left(\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) \right),$$

where the condition

$$\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) > 0$$

is guaranteed by (3.18) (or (1.4)). Finally, (5.12) is proven after multiplying and dividing $(n!)^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right)}$ on the inside of the log in (5.14). $\square$

Lemma 5.6. *For any $k, \theta > 0$, there exists a constant $c_1 = c_1(\alpha, \mathcal{M}, k, \theta) > 0$ such that, by setting $n_t = \lfloor c_1 t \rfloor$, it holds that*

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log(k^{n_t} \theta^{n_t/2} U_{n_t}(t)) \geq \left(k\sqrt{\theta} \right)^{\frac{2a}{2ab - \alpha b + 2ar}} \left(\rho^{1/2} \right)^{\frac{2a}{2ab - \alpha b + 2ar}}. \quad (5.15) \quad \text{E:69ineq}$$

Proof. Let $\epsilon > 0$. Lemma 5.5 guarantees the existence of an $N_\epsilon > 0$ so for all $n > N_\epsilon$

$$(n!)^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right)} U_n(1) \geq \exp(n(\log(R) - \epsilon)) = R^n e^{-n\epsilon}$$

where

$$R = \frac{\rho^{1/2}}{\left(\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) \right)^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right)}}.$$

Now fix a $c > 0$ and let $n_t := \lfloor ct \rfloor$. Notice that $n_t \geq N_\epsilon$ for any $t > t_\epsilon := (N_\epsilon + 1)/c$.

Let $t > t_\epsilon$.

$$k^{n_t} \theta^{n_t/2} U_{n_t}(t) = k^{n_t} \theta^{n_t/2} t^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) n_t} U_{n_t}(1) \geq \frac{k^{n_t} \theta^{n_t/2} t^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) n_t}}{(n_t!)^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right)}} R^{n_t} e^{-n_t \epsilon}$$

Notice that $\lfloor ct \rfloor / t \rightarrow c$ as $t \rightarrow \infty$ which means that $n_t / t \rightarrow c$ as $t \rightarrow \infty$. With this we can say

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) = c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) =: I(n_t)$$

Now,

$$\begin{aligned} I(n_t) &\geq c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left((kR\sqrt{\theta})^{n_t} \frac{t^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) n_t}}{(n_t!)^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right)}} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) + c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left[\left(\frac{t}{n_t} \right)^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) n_t} \frac{n_t^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) n_t}}{(n_t!)^{\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right)}} \right] - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(\frac{n_t^{n_t}}{(n_t)!} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) - c\epsilon \end{aligned}$$

and letting ϵ tend to 0 we see that

$$I(n_t) \geq c \left[\log(k\sqrt{\theta}R) - \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) \log(c) + \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b} r \right) \right] =: h(c).$$

In order to maximize $h(c)$, notice that

$$h'(c) = 0 \iff c^* = (k\sqrt{\theta}R)^{\frac{a}{b(a-\frac{a}{2}+\frac{a}{b}r)}}.$$

After plugging c^* and replacing R we arrive at the following inequality

$$I(n_t) \geq (k\sqrt{\theta})^{\frac{a}{b(a-\frac{a}{2}+\frac{a}{b}r)}} (\rho^{1/2})^{\frac{a}{b(a-\frac{a}{2}+\frac{a}{b}r)}},$$

which proves (5.15) after some simplification. $\square$

We are now ready to prove (5.1).

Proof of (5.1). By proposition 5.3, for and $p, q > 0$ with $p^{-1} + q^{-1} = 1$ we have that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \right|.$$

We now take the supremum over all $f \in \mathcal{H}_+$ with $\|f\|_{\mathcal{H}} = k > 0$. Recall that $W_n(t, \phi) = k^n W_n(t, \phi/k)$ for $\phi \in L^2(\mu)$ and non-negativity of $W_n(t, \cdot)$ on $\mathcal{H}_+$. Let $c > 0$.

$$\begin{aligned} \|u(t, 0)\|_p &\geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=k} \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\ &= \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \sum_{n \geq 0} k^n \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\ &\geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f) \end{aligned}$$

where $n_t = \lfloor ct_p^\beta \rfloor$. Now by choosing c as in Lemma 5.6 we get that

$$\begin{aligned} \liminf_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p &\geq \liminf_{n \rightarrow \infty} \left(\frac{-\frac{1}{2} t_p^\beta k^2}{t_p^\beta} + \frac{1}{t_p^\beta} \log \left[\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f) \right] \right) \\ &= -\frac{1}{2} k^2 + k^{\frac{2a}{2ab-\alpha b+2ar}} (\rho\theta)^{\frac{a}{2ab-\alpha b+2ar}} =: h(k). \end{aligned}$$

By maximizing h for $k > 0$, we see that h is maximized at the point

$$k^* = \left(\frac{2ab - \alpha b + 2ar}{2aB} \right)^{\frac{2ab - \alpha b + 2ar}{2a - 2[2ab - \alpha b + 2ar]}} \quad \text{with } B = (\theta\rho)^{\frac{a}{2ab - \alpha b + 2ar}}.$$

Inserting k^* into h gives us that

$$h(k^*) = B^\beta \left(\frac{2a}{2ab - \alpha b + 2ar} \right)^\beta \left(\frac{2ab - \alpha b + 2ar - a}{2a} \right)$$

and plugging in the value for B proves (5.1). $\square$

6 Appendix

S:Appendix

Proof of Lemma 2.4. Recall that we define $\varphi(x) = C_{\alpha,d}|x|^{-(d-\alpha)}$. We begin by applying the substitution $x' = (\frac{\nu}{2})^{1/a}$ to ρ :

$$\begin{aligned}\rho &= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx) \\ &= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \left(\frac{\nu}{2}\right)^{(d-\alpha)/a} \varphi(x) \left(\frac{\nu}{2}\right)^{-d/a} dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \varphi(x) dx\end{aligned}$$

Now define $f^*(x) = \left(\frac{\nu}{2}\right)^{\frac{-d}{2a}} f\left(\left(\frac{\nu}{2}\right)^{-1/a}x\right)$. Then with this we see that

$$\rho = \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y)f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx$$

and note that

$$\int_{\mathbb{R}^d} f(x)^2 dx = \int_{\mathbb{R}^d} f^*(x)^2 dx$$

and thus

$$\left\{ f \in L^2(\mathbb{R}^d) : \|f\|_2 = 1 \right\} = \left\{ f \in L^2(\mathbb{R}^d) : \|f^*\|_2 = 1 \right\}$$

so we can say that

$$\begin{aligned}\rho &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y)f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f^*\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y)f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \rho_{2,a}\end{aligned}$$

and (2.16) is proven by applying (2.13).

To prove (2.17), we first notice that by applying the substitution $\xi'_{\sigma(j)} = \left(\frac{\nu}{2}\right)^{1/a} \xi_{\sigma(j)}$ that

$$\begin{aligned}& \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}) \\ &= \left(\frac{\nu}{2}\right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi})\end{aligned}$$

and now we can directly calculate

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \left(\frac{\nu}{2} \right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\vec{\xi}) \right] \\
&= \log \left[\left(\frac{\nu}{2} \right)^{-\alpha/a} \right] + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\vec{\xi}) \right]
\end{aligned}$$

and because of both Theorem 1.5 [3] and (2.16) above,

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right) \\
&= \log \left(\left(\frac{\nu}{2} \right)^{-\alpha/a} \right) + \log(\rho_{2,a}) = \log(\rho_{\nu,a}).
\end{aligned}$$

This completes the proof of Lemma 2.4. $\square$

Proof of (3.8). Because

$$\mathcal{F}\widetilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \mathcal{F}f_n(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)}),$$

by the linearity, it suffices to just look at the $\mathcal{F}f(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)})$ terms individually. For any $c > 0$, by (2.9), we have that

$$\begin{aligned}
\mathcal{F}f_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) &= \int_{[0, ct]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(t_{k+1} - t_k, \cdot)} \left(\sum_{j=1}^k \xi_j \right) d\vec{t} \\
&= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(c(t_{k+1} - t_k), \cdot)} \left(\sum_{j=1}^k \xi_j \right) c^n d\vec{t}
\end{aligned}$$

where we applied the substitution $t'_i = t_i/c$. By the scaling property of $\mathcal{F}G$ we see that

$$\begin{aligned}
\int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(c(t_{k+1} - t_k), \cdot)} \left(\sum_{j=1}^k \xi_j \right) c^n d\vec{t} &= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{c^{b+r-1} \mathcal{F}G(t_{k+1} - t_k, \cdot)} \left(c^{b/a} \sum_{j=1}^k \xi_j \right) c^n d\vec{t} \\
&= c^{n(b+r)} \mathcal{F}f_n(\cdot, 0, t)(c^{b/a} \xi_1, \dots, c^{b/a} \xi_n).
\end{aligned}$$

Now by the above scaling property, we can obtain (3.8) as follows:

$$\begin{aligned}
\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \int_0^\infty e^{-t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\widetilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\
&= \int_0^\infty e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\widetilde{f}_n(\cdot, 0; 2t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) 2dt
\end{aligned}$$

$$\begin{aligned}
&= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \widetilde{\mathcal{F}f_n}(\cdot, 0, t)(2^{b/a}\xi_1, \dots, 2^{b/a}\xi_n) \right|^2 \mu(d\vec{\xi}) dt \\
&= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \widetilde{\mathcal{F}f_n}(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 2^{\frac{-nbd}{a}} 2^{\frac{nb(d-\alpha)}{a}} \mu(d\vec{\xi}) dt \\
&= 2^{n(2(b+r)-b\alpha/a)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \widetilde{\mathcal{F}f_n}(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\
&= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt.
\end{aligned}$$

proves (3.8). □

References

- BCC21 [1] Balan, R. M., Chen, L., and Chen, X. (2021). Exact asymptotics of the stochastic wave equation with time-independent noise. *Ann. Inst. Henri Poincaré Probab. Stat.*, to appear.
- BS17 [2] Balan, R. M. and Song, J. (2017). Hyperbolic Anderson Model with space-time homogeneous Gaussian noise. *ALEA, Latin Amer. J. Probab. Math. Stat.* **14**, 799–849.
- BCR09 [3] Bass, R., Chen, X., and Rosen, M. (2009). Large deviations for Riesz potentials of additive processes. *Ann. Inst. Henri Poincaré Probab. Stat.* **45**, 626–666.
- CHHH17 [4] Chen, L., Hu, Y., Hu, G., and Huang, J. (2017). Stochastic time-fractional diffusion equations on $\mathbb{R}^d$. *Stochastics* **89** (1) 171–206.
- CHN19 [5] Chen, L., Hu, Y., and Nualart, D. (2019). Nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$. *Stochastic Process. Appl.* **129**, 5073–5112.
- Chen17AIHP [6] Chen, X. (2017). Moment asymptotics for parabolic Anderson equation with fractional time-space noise: in Skorohod regime. *Ann. Inst. Henri Poincaré: Prob. Stat.* **53** 819–841.
- Chen19AIHP [7] Chen X. (2019). Parabolic Anderson model with rough or critical Gaussian noise. *Ann. Inst. Henri Poincaré Probab. Stat.* **55**, 941–976.
- inicourse09 [8] Dalang, R., Khoshnevisan, D., Mueller, C., Nualart, D., and Xiao, Y. (2009). *A minicourse on stochastic partial differential equations*. Springer-Verlag, Berlin, 2009. xii+216 pp.
- Hu01Heat [9] Hu, Y. (2001). Heat equations with fractional white noise potentials. *Appl. Math. Optim.* **43**, 221–243.
- LiebLoss01 [10] Lieb, E. and Loss, M. (2001). *Analysis*. Second edition. Graduate Studies in Mathematics, 14. American Mathematical Society, Providence, RI.
- nualart18Book [11] Nualart D. and Nualart E. (2018). *Introduction to Malliavin Calculus*. Cambridge University Press.